ASSIGNMENT-12

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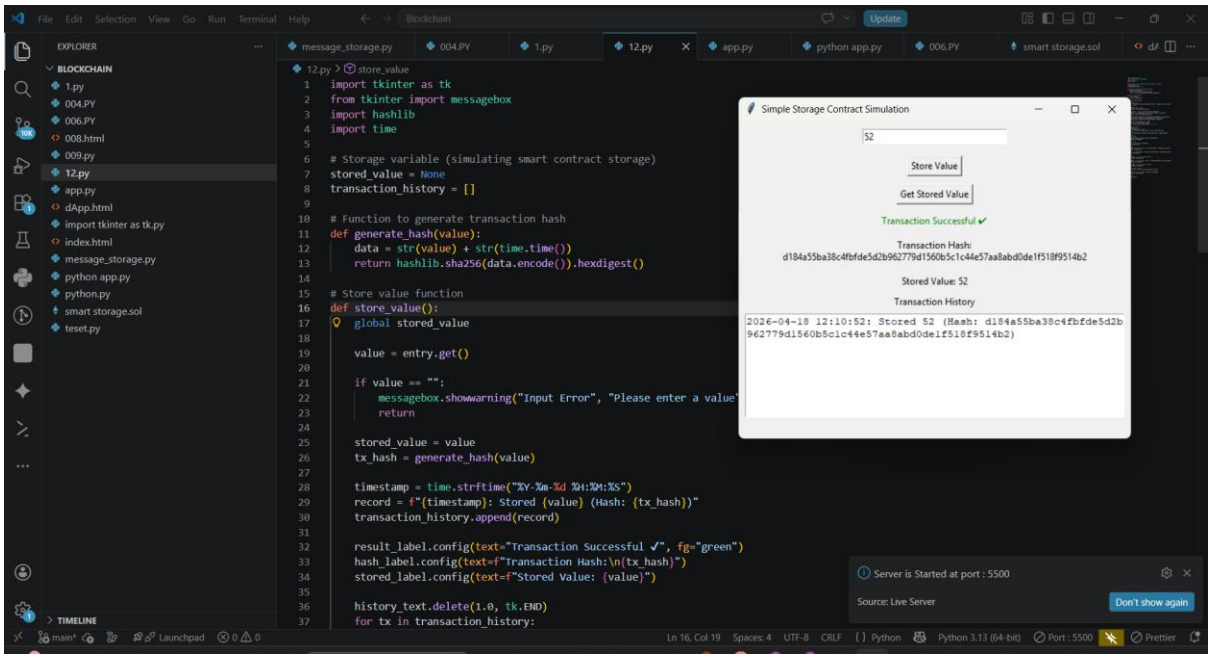
BATCH:29

CODE:

```
import tkinter as tk
from tkinter import messagebox
import hashlib
import time
stored_value = None
transaction_history = []
def generate_hash(value):
    data = str(value) + str(time.time())
    return hashlib.sha256(data.encode()).hexdigest()
def store_value():
    global stored_value
    value = entry.get()
    if value == "":
        messagebox.showwarning("Input Error", "Please enter a value")
        return
    stored_value = value
    tx_hash = generate_hash(value)
    timestamp = time.strftime("%Y-%m-%d %H:%M:%S")
    record = f"{timestamp}: Stored {value} (Hash: {tx_hash})"
    transaction_history.append(record)
```

```
    result_label.config(text="Transaction Successful ✓", fg="green")
    hash_label.config(text=f"Transaction Hash:\n{tx_hash}")
    stored_label.config(text=f"Stored Value: {value}")
    history_text.delete(1.0, tk.END)
    for tx in transaction_history:
        history_text.insert(tk.END, tx + "\n")
def get_value():
    if stored_value is None:
        messagebox.showinfo("Info", "No value stored yet")
    else:
        stored_label.config(text=f"Stored Value: {stored_value}")
root = tk.Tk()
root.title("Simple Storage Contract Simulation")
root.geometry("500x400")
entry = tk.Entry(root, width=30)
entry.pack(pady=10)
store_btn = tk.Button(root, text="Store Value", command=store_value)
store_btn.pack(pady=5)
get_btn = tk.Button(root, text="Get Stored Value", command=get_value)
get_btn.pack(pady=5)
result_label = tk.Label(root, text="")
result_label.pack(pady=5)
hash_label = tk.Label(root, text="", wraplength=400, justify="center")
hash_label.pack(pady=5)
```

```
stored_label = tk.Label(root, text="")
stored_label.pack(pady=5)
tk.Label(root, text="Transaction History").pack()
history_text = tk.Text(root, height=8, width=60)
history_text.pack(pady=5)
root.mainloop()
```

OUTPUT: